

ATPA Assessment – October to December 2024

General Information for Candidates

This examination has 6 tasks numbered 1 through 6 with a total of 40 points. The points for each task are indicated at the beginning of the task.

Each task pertains to the business problem and related data files and data dictionary. There is not an Rmd file that accompanies this exam. Unless otherwise specified, each task builds upon the work and conclusions from prior tasks.

The responses to each specific task should be written in the provided Word template. Where code, tables, or graphs from your own work are required, they should be copied and pasted into the Word template. Instructions for pasting tables are included at the end of this document.

You may use resources such as textbooks and the internet. You may use any analytics software you wish to perform the analysis directed by the tasks. You may not consult with other individuals about the specific business problem, data, and tasks.

Each task will be graded on the quality of your thought process (as documented in your submission), conclusions, and quality of the presentation. Each response should be confined to the prompt as set and written for the audience specified in the prompt. If no audience is specified, assume a technical audience familiar with predictive modeling. In tasks 1-5, various portions of a technical report are to be written but these are not intended to make up an entire report, e.g., a statement of the business problem is not asked for in these tasks. Only write the sections requested.

Your completed Word template should be the only file submitted and will be the only file graded. If any part of your exam was answered in French, include “French” in the file name.

Business Problem

ABCMart, a nationwide grocery store chain, has used a loyalty program for several years to enhance customer retention and increase sales. Despite a significant initial sign-up rate, ABCMart has observed a worrying trend: a substantial number of loyalty program members have churned, meaning they have become inactive, no longer recording purchases from the store. Understanding customer behavior and predicting churn is critical for ABCMart to develop targeted marketing strategies and improve customer satisfaction.

ABCMart's management has tasked your data analytics consulting firm with identifying key factors that influence customer churn within their loyalty program. They are particularly interested in understanding how various customer demographics, shopping behaviors, loyalty tiers, and transaction patterns contribute to churn. Additionally, they seek insights into how payment methods and the use of coupons or promotions impact customer retention.

Your consulting firm is well-equipped to tackle this challenge by leveraging advanced data analysis and predictive modeling techniques. By analyzing the provided datasets, you aim to uncover actionable insights that will help ABCMart reduce churn rates, enhance customer loyalty, and increase profitability.

ABCMart sent unprotected data files to your assistant in an unencrypted email. Noticing the email addresses were directly in the file, your assistant quickly replaced these emails with a specific encryption, so individuals could not be identified directly from the email. Your assistant noted there were potentially some other sensitive features about the data.

The datasets provided include detailed customer profiles, transaction histories, and loyalty program information. These datasets contain a mix of useful variables, including continuous variables like annual income and categorical variables like preferred payment methods. One of the main challenges is to account for potential non-linear relationships between these variables and customer churn. For example, high-income customers might have different shopping patterns compared to low-income customers, affecting their likelihood to churn.

In the project brief, ABCMart's management has outlined several key questions they need you to address:

- What customer characteristics are associated with higher or lower churn rates?
- How do shopping behaviors and transaction patterns influence customer churn?
- What role do loyalty tiers play in customer retention?

ABCMart expects both a technical report and an executive summary. While the management team has some familiarity with data analytics concepts, they value clear, actionable insights over technical jargon. The most crucial part of your communication will be an executive summary with clear recommendations that ABCMart can use to implement strategies to reduce customer churn.

File List

- Two CSV files:
 - customer_data.csv: Contains customer demographic information and loyalty program details. Includes the churn variable that is used to determine if a customer has discontinued shopping at the store.
 - transaction_data.csv: Contains transaction history with detailed purchase and payment information.
- One Excel file called DataDictionary.xlsx: Contains the data dictionary for the above files.

By delivering this project, you will help ABCMart make informed decisions to enhance customer loyalty and reduce churn. Your detailed statistical analyses, visualizations, and report will provide valuable insights that drive effective marketing strategies and operational improvements.

Task 1 (10 points)

Perform the following data preparation tasks:

a) Clean and prepare the data for analysis:

- Create one combined dataset by appropriately joining the tables from `customer_data.csv` and `transaction_data.csv`. Describe and justify the types of joins used. Consider that some customers may not have transaction records, and some transactions may not match existing customers due to data inconsistencies.
- The target variable is Churn in the `customer_data.csv` file. Some customer records may have missing values in critical fields such as AnnualIncome or Age. Briefly describe approaches to handling records with missing values, recommend, and implement one approach.
- Analyze the predictor variables provided. Based on your judgment before fitting any models, remove variables with no predictive value or that may introduce bias. Explain your choices.
- Check that the variables have values that make sense and are appropriate for modeling.
- Ensure internal consistency between variables that should contain related information (e.g., PreferredPaymentMethod in customer data vs. actual PaymentMethod used in transactions).

b) Feature Engineering:

- Create the following new variables that may enhance the predictive power of your models.
 - a) how frequently someone shops
 - b) amount of money they spend per visit

c) Exploratory Data Analysis:

- Explore the distribution of the target variable Churn.
- Visualize no more than two relationships between the target variable and other predictors that you believe might be significant.

Write a brief section of the technical report explaining your data preparation work. Since your code or the transformed data itself will not be available to the reader, all evidence of the data preparation tasks must be contained in your report. This may include written descriptions as well as charts. Be sure that evidence of dealing with all the steps above is included, including the validation checks, where you should describe the checks performed and report on the results.

Task 2 (2 points)

ABCMart has allowed you access to their data. However, they have not taken any measures to account for any security or privacy of the data. They simply sent unprotected data over in an unencrypted email. Based on this information and from inspecting the data, answer the following two questions:

- c) In the context of the Actuarial Standards of Practice, what would your responsibility be now concerning this data? Point to specific language in the ASOPs to support this.
- d) What standards of practice would ABCMart have violated if they adhered to actuarial standards of practice? Point to specific language in the ASOPs to support this.

Task 3 (8 points)

Perform the following modeling tasks:

- a) Using the data prepared in Task 1, create datasets for model training and testing. Perform reasonability checks to assess if the data split is appropriate.
- b) Fit a logistic regression model for the target Churn. Outline and justify your variable selection approach and final variables included.
- c) Fit a random intercept model for the target Churn using Region as a random effect. You may use input variables as selected from the linear model. Explain why region may be a good choice for the random variable in this data set. Explain why other categorical variables would not have been good choices.
- d) Choose two performance measures for this model's predictions and apply on both the training and test datasets. Evaluate these measures for goodness of fit and over fitting.

Write a section of the technical report that covers all the work above. Identify and discuss the most important predictor variables for the logistic regression model and the mixed model. In the report, discuss the choice of random and fixed predictors. State what advantages the mixed model may have. Report on the goodness of fit of the models.

Task 4 (8 points)

Using the same training and test data from Task 3, perform the following modeling tasks:

- a) Fit a Neural Networks model to the data.
- b) Outline five hyper-parameters that can be used for tuning and fitting a neural network model.
- c) Apply model tuning to at least three hyper-parameters of the neural network model to improve model fit. Try to choose hyperparameters that will make the model fit better. Justify your choices.
- d) Use the same metric as Task 3 and evaluate the neural network model. Check for goodness of fit and for overfitting.

Write a section of the technical report that covers all the modeling work above for the Neural Network model. Include a discussion of the model tuning approach used and justify your choice of final hyper-parameter values. In the report, discuss how this model performs against the model in Task 3.

Task 5 (6 points)

Examine variables that may be important for churn using shapley values.

- a) Fit a random forest model to the data. Tune the model by finding a good complexity parameter and/or tree depth. Check for over fitting by evaluating accuracy on the training and test set.
- b) Choose 5 individuals who have churned and 5 individuals who have not. Find the shapley values for these observations. Interpret the shapley values in the context of this problem.

Write a section of the technical report discussing your modeling work and addressing these requests.

Task 6 (6 points)

Write a one-to-two-page executive summary to ABCMart's management team that summarizes the information they need to make well-informed decisions on strategies to reduce customer churn. This will incorporate some of the work and findings from prior tasks and will also include additional sections such as:

- **Statement of the Business Problem:** A clear and concise description of the challenge ABCMart is facing.
- **Key Findings:** Highlight the most significant factors influencing customer churn.
- **Recommendations:** Provide actionable suggestions on how ABCMart can reduce churn based on your analysis.
- **Limitations:** Briefly mention any limitations of your analysis that the management should be aware of.

Ensure that the executive summary is written in clear, non-technical language suitable for an audience without a data science background.

Important Notes about Inserting Tables

If you need to insert an Excel table into your document, then you must follow the instructions below to copy and paste the table from Excel into your Word document.

Do:

- Ensure your table fits in the page margins by:
 - Selecting the table
 - Right-clicking on the table to view the pop-up menu
 - Hover your mouse over “Auto Fit”
 - Click “AutoFit to Window”.

Do **not**:

- Paste as Picture
- Insert Object
- Use Word’s “Insert, Table, Excel Spreadsheet” function, since this will not paste your table in the correct format.

Pasting a **table** as a picture or as an object will result in an automatic disqualification of your submission. (**Graphs** pasted as pictures are **not** affected by this requirement.)

Steps for pasting tables (for Windows versions of Word/Excel):

1. Copy the cells from your Excel spreadsheet. If you use the Copy command on the Home tab, do **not** select the option to “copy as picture”.
2. In your Word document, turn on the “Show/Hide Paragraph Marks” feature.
3. In your Word document, right-click where you want to insert your table.

4. In the menu that pops up, under Paste Options, select any of the first four options:
 - a. “Keep Source Formatting”
 - b. “Use Destination Styles”
 - c. “Link & Keep Source Formatting”
 - d. “Link & Use Destination Styles”
5. If you have “Show/Hide Paragraph Marks” turned on, you should see small circles at the end of each cell in your table. This is how you can know whether or not you have pasted your table correctly.

Steps for pasting tables (for Mac versions of Word/Excel):

1. Copy the cells from your Excel spreadsheet. If you use the Copy command on the Home tab, do **not** select the option to “copy as picture”.
2. In your Word document, turn on the “Show All Nonprinting Characters” feature.
3. In your Word document, right-click where you want to insert your table.
4. In the menu that pops up, click Paste (**not** Paste Special).
5. If you have “Show All Nonprinting Characters” turned on, you should see small circles at the end of each cell in your table. This is how you can know whether or not you have pasted your table correctly.

As You Complete Your Tasks...

Your name should not appear anywhere in your submission, or in the filename, or in any way identify you as the author. If you believe you should refer to yourself in the context of an answer, use “ATPA Candidate” instead of your actual name.